
例題 4.10 つぎの s 関数のラプラス逆変換を求めよ。

(1) $\frac{1}{s^2(s+2)}$

(2) $\frac{1}{s(s+1)^3}$

(3) $\frac{s-1}{s^2-2s+2}$

例題 5.1 つぎの微分方程式の解を求めよ。

(1) $x' + 2x = e^{-t}, \quad x(0) = 3$

(2) $x'' + 6x' + 10x = 0, \quad x(0) = x'(0) = 2$

例題 5.2 つぎの微分方程式の解を求めよ。

(1) $x'' - 3x' + 2x = 0, \quad x(0) = 0, \quad x(1) = 1$

(2) $x'' + x = e^{-t} \quad (0 < t < 1), \quad x(0) = 0, \quad x(\pi/2) = 0$

例題 5.5 つぎの積分方程式の解を求めよ。

$$f(t) - 2 \int_0^t e^{-3(t-\tau)} f(\tau) d\tau = e^{-t}$$

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P85. 13) 4.10

$$(1) \mathcal{L}^{-1} \left[\frac{1}{s^2(s+2)} \right] = \mathcal{L}^{-1} \left[\frac{\frac{1}{2}}{s^2} \right] + \mathcal{L}^{-1} \left[\frac{\frac{1}{4}}{s+2} \right] + \mathcal{L}^{-1} \left[\frac{-\frac{1}{4}}{s} \right]$$

$$= \frac{1}{4} e^{-2t} + \frac{1}{2} t - \frac{1}{4}$$

$$(2) \mathcal{L}^{-1} \left[\frac{1}{s(s+1)^3} \right] = \mathcal{L}^{-1} \left[\frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2} + \frac{D}{(s+1)^3} \right]$$

$$A (\text{cover } s) = A = \frac{1}{(s+1)^3} \Big|_{s=-1} = 1$$

$$B, C, D (\text{formula}) = B = \lim_{s \rightarrow -1} \frac{1}{s} = -1$$

$$C = \lim_{s \rightarrow -1} \frac{1}{s^2} = -1$$

$$D = \lim_{s \rightarrow -1} \frac{1}{s} = -1$$

$$(3) \mathcal{L}^{-1} \left[\frac{s-1}{s^2-2s+2} \right] = \mathcal{L}^{-1} \left[\frac{s-1}{(s-1)^2+1} \right]$$

$$= e^t \cos t.$$

P91. 13) 5.1

(1) Convolution.

$$\mathcal{L}[x'] + \mathcal{L}[2x] = \mathcal{L}[e^{-t}]$$

$$s\mathcal{L}[x] + 2\mathcal{L}[x] - x(0) = \frac{1}{s+1}$$

$$\mathcal{L}[x] = \frac{\frac{1}{s+1} + x(0)}{s+2}$$

$$\mathcal{L}^{-1}[\mathcal{L}[x]] = \mathcal{L}^{-1} \left[\frac{\frac{1}{s+2} + \frac{1}{s+1}}{s+2} \right]$$

$$= 2e^{-2t} + e^{-t}$$

(2) Convolution.

$$\mathcal{L}[x''] + 6\mathcal{L}[x'] + 10\mathcal{L}[x] = 0$$

$$s^2\mathcal{L}[x] - sx(0) - x'(0) + 6[s\mathcal{L}[x] - x(0)] + 10\mathcal{L}[x] = 0$$

$$\mathcal{L}[x] = \frac{2s+4}{s^2+6s+10} = 2 \frac{s+3}{(s+3)^2+1} + \frac{8}{(s+3)^2+1}$$

$$\mathcal{L}^{-1}[\mathcal{L}[x]] = 2e^{-3t} \cos t + 8e^{-3t} \sin t$$

P93. 13) 5.2

(1) Convolution.

$$s^2\mathcal{L}[x] - sx(0) - x'(0) - 3s\mathcal{L}[x] - 3x(0)$$

$$+ 2\mathcal{L}[x] = 0 \quad \begin{cases} x(0) = 0 \\ x'(0) = 1 \end{cases}$$

$$\mathcal{L}[x] = \frac{C}{s^2-3s+2}$$

$$\mathcal{L}^{-1}[\mathcal{L}[x]] = C(e^{2t} - e^t)$$

$$x(1) = 1 \quad \mathcal{L}[1] = \frac{1}{s}$$

$$x(t) = \frac{e^{2t} - e^t}{e^2 - e}$$

P99. 13) 5.5

Convolution. Theorem.

$$f(t) = 2 \int_0^t e^{-3(t-r)} f(r) dr = e^{-t}$$

$$\mathcal{L}[f(t)] = 2\mathcal{L}[e^{-3t}] \mathcal{L}[f(t)] = \frac{1}{s+1}$$

$$\mathcal{L}[f(t)] = \frac{s+3}{(s+1)^2}$$

$$f(t) = (2t+1)e^{-t}$$